

Nmap

Category: Reconnaissance • Network Discovery • Security Auditing

Purpose: A practical reference for learning Nmap, performing authorized network discovery and port scanning, interpreting results, and understanding common Nmap features.

1. What is Nmap?

Nmap (Network Mapper) is an open-source tool for network exploration and security auditing. It can discover available hosts, ports, services and versions, make operating-system guesses, and provide other target information. ■cite■turn0search2■

2. Where Nmap Fits in Cybersecurity

Area	Use
Reconnaissance	Discover hosts, ports, services, and other target information.
Security auditing	Support authorized assessments and network inventories.
Defensive validation	Check whether exposed services match the intended system configuration.
SOC / monitoring	Help identify services that may require monitoring or restriction.

3. Basic Syntax

```
nmap [Scan Type] [Options] <target>
```

Targets may be specified using IP addresses, hostnames, or other supported target specifications. ■cite■turn0search2■

4. Basic Scanning

```
nmap 192.168.13.10
```

Basic scan of a single lab host.

```
nmap -p 22,80,443 192.168.13.10
```

Scan selected TCP ports.

```
nmap -p- 192.168.13.10
```

Scan all TCP ports from 1–65535.

5. Port States

State	Meaning
open	An application is listening on the port.
closed	The port is reachable, but no application is listening.
filtered	A firewall/filter prevents Nmap from determining whether it is open or closed.
unfiltered	The port responds, but Nmap cannot determine whether it is open or closed.
open filtered	Nmap cannot determine whether the port is open or filtered.

6. Service & Version Detection

A port number does not guarantee which service is actually running. Nmap can interrogate discovered ports to identify the protocol, application name, version, and sometimes additional information. ■cite■turn0search0■turn0search10■

```
nmap -sV 192.168.13.10
```

-sV enables service and version detection.

```
nmap -sV --version-light 192.168.13.10
```

Uses lower version-detection intensity for faster scanning. ■cite■turn0search0■

7. OS Detection

```
nmap -O 192.168.13.10
```

Attempts to identify the target operating system. Treat fingerprint-based results as estimates rather than absolute proof.

8. Nmap Scripting Engine (NSE)

NSE extends Nmap using Lua scripts for tasks such as network discovery, enhanced version detection, vulnerability detection, and other specialized assessment tasks. Some scripts can be intrusive, so use them only on authorized targets.

```
nmap -sC 192.168.13.10
```

Runs the default NSE script set.

```
nmap --script <script-name> TARGET
```

Runs a selected NSE script.

9. Useful Scan Options

Command	Purpose
<code>nmap -sV TARGET</code>	Service and version detection.
<code>nmap -O TARGET</code>	OS detection.
<code>nmap -sC TARGET</code>	Default NSE scripts.
<code>nmap -sV -sC TARGET</code>	Version detection plus default scripts.
<code>nmap -A TARGET</code>	Enables OS/version detection, script scanning, traceroute, and related functionality.
<code>nmap -p 1-1000 TARGET</code>	Scan a defined port range.
<code>nmap -oN scan.txt TARGET</code>	Save normal-format output to a file.

10. Reading Nmap Output

Field	Meaning
PORT	Port number and transport protocol, e.g. 22/tcp.
STATE	Port state such as open, closed, or filtered.
SERVICE	Service associated with the port.
VERSION	Application/version information when version detection succeeds.

Example: 22/tcp open ssh OpenSSH indicates TCP port 22 is open and Nmap identified SSH.

11. Practical Lab Workflow

- 1. Choose an authorized target such as a VM in your VirtualBox lab.
- 2. Run a basic scan to discover reachable ports.
- 3. Use -sV to identify services and versions.
- 4. Use -O when OS fingerprinting is appropriate.
- 5. Use selected NSE scripts only after understanding their purpose.
- 6. Save results and compare them with the services actually configured on the lab machine.
- 7. Investigate unexpected exposed services from a defensive perspective.

12. Security Perspective

Nmap is primarily a discovery and assessment tool. Its results can reveal the visible attack surface of a host by showing reachable ports and services. Defenders can compare these findings with intended configuration and security policy.

13. Common Mistakes

- Assuming the service is guaranteed by its port number.

- Treating OS or service identification as infallible.
- Scanning systems without authorization.
- Running intrusive NSE scripts without understanding their behavior.
- Ignoring UDP services when relevant to an assessment.
- Looking only at open ports and ignoring filtered or closed results.

14. Quick Revision

Concept	Remember
Nmap	Network exploration and security auditing tool.
Port scan	Determines the state of ports on a target.
-sV	Service and version detection.
-O	OS detection.
-sC	Default NSE scripts.
-A	Broader scan including OS/version detection, scripts, and traceroute.
NSE	Lua-based scripting framework extending Nmap.

15. Key Takeaways

- Nmap is widely used for network discovery and security auditing.
- Open, closed, and filtered states provide different information about network exposure.
- Service/version detection helps determine what is running behind a discovered port.
- NSE extends Nmap with scripts for discovery and security assessment.
- Use Nmap only against systems and networks you are authorized to assess.

Reference: Official Nmap Reference Guide and Nmap Network Scanning. [■cite■turn0search1■turn0search2■](#)